

RaceDay API Endpoint Plan

Module: Programming 2B (PROG6212) | Student: ST10458180 | Part 1 – System Planning, Section B

Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Creates a new user account as either an Organiser or a Participant.	None (public)	{ fullName, email, password, role }	201 Created – new user id and role. 400 Bad Request – missing fields or email already in use.
POST	/api/auth/login	Validates a user's email and password and returns a JWT so they can access protected endpoints.	None (public)	{ email, password }	200 OK – JWT token and role. 401 Unauthorized – invalid credentials.

User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Returns the profile of the currently logged in user (Organiser or Participant).	Any (logged in)	None	200 OK – user profile fields. 401 Unauthorized – no/invalid token.
PUT	/api/users/me	Lets the logged in user update their own profile information.	Any (logged in)	{ fullName, email }	200 OK – updated profile. 400 Bad Request – invalid data.

Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Returns a list of all upcoming events so anyone can browse them.	Any	None	200 OK – array of events.
GET	/api/events/{id}	Returns the full detail of a single event.	Any	None	200 OK – event detail. 404 Not Found – event does not exist.
POST	/api/events	Allows an Organiser to create a new event.	Organiser	{ eventName, description, eventDate, location, distanceKm, eventType }	201 Created – the new event. 400 Bad Request – missing/invalid fields.
PUT	/api/events/{id}	Allows an Organiser to update an event they created.	Organiser	{ eventName, description, eventDate, location, distanceKm, eventType }	200 OK – updated event. 403 Forbidden – not the owner. 404 Not Found.
DELETE	/api/events/{id}	Allows an Organiser to delete an event they created.	Organiser	None	204 No Content – deleted. 403 Forbidden – not the owner. 404 Not Found.

Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Returns all categories available for a specific event (e.g. Under 20, Senior, 10km).	Any	None	200 OK – array of categories. 404 Not Found – event does not exist.
POST	/api/events/{eventId}/categories	Allows an Organiser to add a new age/distance category to their event.	Organiser	{ categoryName, minAge, maxAge }	201 Created – the new category. 400 Bad Request – invalid data.
PUT	/api/categories/{id}	Allows an Organiser to update a category on their own event.	Organiser	{ categoryName, minAge, maxAge }	200 OK – updated category. 403 Forbidden. 404 Not Found.
DELETE	/api/categories/{id}	Allows an Organiser to remove a category from their own event.	Organiser	None	204 No Content. 403 Forbidden. 404 Not Found.

Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{eventId}/enrolments	Lets a Participant enter an event under a chosen category. Links the participant, the event and the category.	Participant	{ categoryId }	201 Created – enrolment record. 404 Not Found – event/category missing. 409 Conflict – already enrolled.
GET	/api/events/{eventId}/enrolments	Lets an Organiser view everyone who has enrolled in their event.	Organiser	None	200 OK – array of enrolments. 404 Not Found – event does not exist.
GET	/api/users/me/enrolments	Lets a Participant view their own enrolment/participation history.	Participant	None	200 OK – array of the participant's enrolments.

Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments/{enrolmentId}/results	Lets an Organiser capture a finish time and finishing position for a participant after the event.	Organiser	{ finishTime, finishPosition }	201 Created – result record. 400 Bad Request – invalid data. 404 Not Found – enrolment does not exist.
GET	/api/events/{eventId}/results	Returns the full result list for an event (e.g. a results board).	Any	None	200 OK – array of results. 404 Not Found – event does not exist.
GET	/api/users/me/results	Lets a Participant view only their own results across all events.	Participant	None	200 OK – array of the participant's results.